

Navigating the RISM data with RISM Online

Andrew Hankinson
Laurent Pugin
andrew.hankinson@rism.digital
laurent.pugin@rism.digital
RISM Digital Center
Bern, Switzerland

ABSTRACT

In 2021, the RISM Digital Center introduced RISM Online. This represented a shift in how we present the RISM data to a global audience, supporting new methods of digital research and keeping the RISM project central to modern music scholarship. RISM Online is designed from the ground-up to move past a simple descriptive catalogue, treating the results of over 70 years of indexing, collating, organizing, and curating musical source descriptions and authorities as a significant and valuable data resource in itself. In this paper we explore the shift from catalogue to dataset more closely, looking at some of the unique and valuable information captured by RISM that can be of use to data-driven musicology. On the way we will identify how RISM Online is making this data available through the tools and access points we have built. Finally, we will report on some ongoing experiments with the RISM data as we seek to exploit the relationships captured therein as an active area of future work.

CCS CONCEPTS

• **Information systems** → **Information retrieval**; **Web services**; **Digital libraries and archives**; • **Applied computing** → **Sound and music computing**.

KEYWORDS

RISM, digital musicology, libraries, digital humanities, linked data, databases, search, music

ACM Reference Format:
Andrew Hankinson and Laurent Pugin. 2024. Navigating the RISM data with RISM Online. In *11th International Conference on Digital Libraries for Musicology (DLfM 2024), June 27, 2024, Stellenbosch, South Africa.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3660570.3660576

1 INTRODUCTION

Since 1952, the Repertoire International des Sources Musicale (RISM) project has been cataloguing and collating music collections around the world. Through the efforts of countless individuals, the RISM catalogue has become an indispensable tool for anyone doing musical source studies. For most of the project's history, the RISM catalogues have served as a "union catalogue" of musical sources from around the world. The RISM union catalogues enabled users to identify the location of the physical sources held in libraries across Europe and North America. Notably, this was also an important social development as it was one of the first post-war international collaborations, particularly between German, French, and British scholars and librarians.

The road to making the RISM data available online has been a long one. Initially, RISM published their catalogues in printed books, organized in series. These books sometimes took years or even decades to issue new volumes in the series or publish errata to older volumes. This started to change through the 1970s and 1980s with Series A/II and the introduction of computerized cataloguing, first at the RISM central office and then to the working groups in other countries. In 1990, the PiKaDo application was provided to cataloguers "in the field" to create their catalogue records. These records were mailed to the central office on floppy disks, and then collated into a single catalogue. In 1995 the A/II records were published on CD-ROM, and updates issued to subscribers annually. In 1997, the first RISM records appeared online[18], representing the entries from the USA. The full A/II catalogue was finally made available, online and for free, in 2010 through a partnership with the Bavarian State Library. In 2016, after several attempts at finding suitable software, the cataloguing process also went fully online with the introduction of Muscat to all working groups[10]. Muscat had been in development since 2002 in a joint project with the UK and Swiss working groups. With Muscat, and with the online catalogue, updates to the RISM catalogue were issued much more frequently, typically monthly. Retro-conversion efforts for printed and handwritten index cards, as well as optical character recognition of some of the printed books, have brought much of the pre-digital efforts of the RISM catalogue into Muscat. However, even now not all entries in the printed catalogues are available online.

Historically, RISM has focused on collections held in European and North American libraries, but in recent years this has grown to include collections from other regions through partnerships with East Asian and South American libraries. RISM also had time-based boundaries for its earliest descriptive efforts, in order to maintain a reasonable size for the printed volumes. With the shift to the direct cataloguing model, and publication of the records online, such restrictions are no longer necessary. Libraries have started to include more modern material in their descriptive efforts, broadening the time periods to the 19th, 20th, and even 21st centuries.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
DLfM 2024, June 27, 2024, Stellenbosch, South Africa
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1720-8/24/06
https://doi.org/10.1145/3660570.3660576

Today, with the collective efforts of countless individuals over seven decades, careful stewardship, and several monumental transitions through both cataloguing systems and publication media, the RISM project is the curator of the largest musical source description database in the world. And yet, the underlying data used to build the RISM catalogue is a significant and mostly un-tapped, resource. Typically the "size" of RISM is quoted in the numbers of sources or authority records, but other numbers, such as the number of triples we have extracted (over 77 million), the search index size in Solr (over 9 gigabytes) and the number of individual search documents in our index (approaching 6 million) provide an alternative, but perhaps more illustrative, measure of the size and richness of the data that underpins the catalogue.

2 THE RISM DATA

The RISM data are the collected set of records captured in the everyday work of RISM cataloguers, represented as data points in distinct fields and subfields in the cataloguing records, describing individual properties of every record. A typical record may contain tens or hundreds of itemized observations and, through the use of the Plaine and Easie Code (PAEC), short extracts of symbolic music notation. Almost all of the RISM data is stored in the MARC21[1] format. This is a well-known and highly optimized format for capturing bibliographic data, in use in most libraries with electronic catalogues.

Several different classes of records are captured in the process of cataloguing sources. Bibliographic records capture the individual manuscript or printed book information. The "authority records" capture information about people, institutions, places, and several other record types that serve to inter-relate bibliographic and authority records. These authorities are, in themselves, a valuable data set as they identify the actors and places that have had notable roles in the production and use of musical works. The structure, contents, and ways each of the record types inter-relate is what we will call the "RISM Data Model."

This section will present some background on the structure and inter-related nature of the RISM data. It will describe how source records are structured, and how the authorities serve to inter-relate and expose the source data in useful ways. We will also briefly describe musical incipits and how they can be used as valuable access points into the RISM data.

2.1 Sources

Source records are at the core of the RISM cataloguing activities, so it is perhaps unsurprising that they are also have the most complex data models and structure. Manuscripts and Prints can be complex bibliographic objects, and the level of description available in "standard" catalogues does not generally supply enough detail that can accurately capture that complexity for the purpose of identification and disambiguation, in large part because some of the limitations inherent in the MARC21 format, as implemented by most library catalogues[17]. As a result, the RISM data model captures source records using a different structure than employed by most library catalogues, and this structure provides our cataloguers with greater expressiveness when describing source records.

[1] https://www.loc.gov/marc/

There are three types of Source record in RISM: Single-Item, Collection, or Item. The Single-Item source record is one where the bibliographic item and its contents are described in a single record. In contrast, a Collection record serves as a "parent" record to one or more Item records, with each record describing an entry in the source. The use of the Item as a separate source record, when it in fact describes the contents of a Source, is a common source of confusion when describing the RISM data model. However, since MARC21 does not permit detailed contents descriptions, the Item record as a standalone record allows for detailed source descriptions of individual components of a Source. Individual entries within a manuscript record can be described as their own bibliographic entity, a technique referred to as "Analytical Cataloguing"[8]. Particularly with manuscripts, individual "sources" may be bound together into a single volume. It is not uncommon for a single manuscript to be compiled across several years, decades, or even centuries and have widely different content bound together within a single codex. The Collection and Item relationship was designed to recognize this variability and to treat the individual entries within the manuscript as a source in its own right.

This is slightly different for prints, where the production process is typically a singular event and the contents more uniform than a manuscript. However, the adoption of the same Collection and Item model maintains consistency across source records. Source records for prints describe all copies of that particular print, but do not describe any single physical copy. Individual copies are described in Holding records, which relates the Source to a holding institution. The Holding record is attached to the Single-Item or Collection record for the print, and not to individual Item records within the Collection.

2.2 Authorities

Authority records are entries that describe a person, an organization, or a place. They serve to bind the sources together and provide additional ways in which to approach the sources by giving different points of entry into the database. Users may be performing exploratory work and do not know the exact source they are looking for, but they may know the composer, publisher, or the holding institution. The authority records allow them to traverse the relationship between what they know, and what they do not know.

Beyond simply capturing a name, RISM authorities can contain additional data on that entity. Birth and death dates, biographies, familial, or professional relationships, roles and jobs, and known places they were associated with are common elements to capture for people. Likewise, institutional addresses, histories, geographical location, and other relationships are common elements that may be captured as part of an organization's authority record. With over 150,000 people and 26,000 organizations, the RISM authorities represent a significant database of historical actors involved in the production and distribution of music through the centuries. Many of these actors are unique to RISM and not present in other library authority files, since RISM records contain more extensive relationship tracings than most library catalogues.

2.3 Sigla

The institutional authorities also serve as the assignment authority for RISM sigla: short codes that uniquely identify an organization that operates as a holding institution for collections. These sigla are widely used in musicological research as a compact and unique method of identifying a holding institution, particularly in citations or concert programs where space is at a premium.

The structure of a RISM siglum consists of a country code[2], followed by a dash, one or more upper-case letters indicating a city, and (typically) one or more lower-case indicators for the institution. Thus the code GB-Lbl indicates Great Britain, London, British Library. Occasionally the institution code is omitted as is the case with D-B (Germany, Staatsbibliothek zu Berlin).

2.4 Incipits

An incipit, in the context of the RISM catalogue, is a short monophonic segment of the musical content of a piece of music, intended to serve as a way of identifying a musical piece using the musical content itself. RISM incipits are encoded using the Plaine and Easie format[3, 7], a compact text-based encoding specifically designed to capture melodic incipits in catalogues. There are over 1.9 million incipits in the RISM catalogue, making it a significant dataset of encoded musical notation.

2.5 Other Data Types

There are many other types of curated authority data in the RISM database—too many to explore in detail here. Secondary literature, liturgical festivals, subject headings, and geographic place authorities are all present and linked to the source and authority records, creating a vast set of relationships in the RISM data.

3 RISM ONLINE

RISM Online[3] is a web application purpose-built to provide a platform for delivery of the RISM data. RISM Online extends the traditional view of RISM as a source catalogue by providing dedicated search interfaces for both sources and authorities, and by allowing users to easily navigate these relationships. In this section we will describe some of the core functions and features of RISM Online, to help orient readers to the use of the resource, and to provide details of the technologies we have developed or adopted in the hopes that they may be useful for others when implementing their own digital tools. Our hope is that this is useful for a wide audience, from those who may be interested in a high-level overview of RISM Online, to those who are seeking details on how it was built and some of the technical design details and processes that were involved in making it. RISM Online is completely Open Source, with a list of the repositories containing code that was built or enhanced in the making of it provided in Appendix A.

3.1 Interaction Overview

The front page of RISM Online has four top-level search interfaces available: Sources, People, Institutions, and Incipits. For simple keyword queries, RISM Online will attempt to match the query against all record types. However, each of these interfaces also supports different facets to allow the user to narrow down the searches to their desired set of results. As the user enters search terms and operates the facets, the search interface will constantly update the number of results that would be retrieved to help the user construct their query in an iterative fashion. When the user narrows the number down to a manageable amount, they submit the search, bringing them to a view where they can navigate the results and perform further updates to their query. Clicking on a search result opens up a record preview, allowing the user to view the record without leaving the search results page. A "full record" link brings the user to a dedicated page for that record.

Many of the full record pages feature a tabbed navigation interface. One tab displays the details of the record. A second tab brings users to a dedicated search interface for source records related to that record. This appears on Collection-type records for sources, and allows users to search within the contents of the source. For institution and person records, the search interface provides the user with the ability to search the source records that are related to that person or institution. This is useful for providing a dedicated search page for sources attached to a particular composer, or for navigating the holdings of an institution. Every authority record that is related to one or more source has this feature.

The incipit search uses the Plaine and Easie Code (PAEC) as its query language. An on-screen piano keyboard allows users to click to enter pitches (with split black keys for entering sharp and flat notes separately). This keyboard interface translates the clicks to PAEC, and writes the results into a text field, allowing users to edit their query before submission. Like the other search interfaces, a preview of the number of results that would be returned is automatically calculated as the user types. Incipit searches may also be combined with keyword queries and filters, allowing users to combine notation-based searches with other metadata taken from the source record. Every incipit displayed in RISM Online has a set of additional functions, including the ability to use any incipit as the basis of a new search, to help users find similar incipits. Incipit search results show highlighted notes, indicating to the user where their matches were found. An in-depth exploration of the implementation behind the incipit search will come later in this paper.

The user interface for RISM Online has translations in seven languages: English, French, German, Italian, Polish, Portugese, and Spanish. The user's language will be automatically detected from their browser settings, and will default to English for languages other than these seven. A language chooser allows users to change the language of the interface "on-the-fly." The language values are shared with the Muscat project, and we have a team of volunteer translators who, through the use of the Crowdin tool[4], provide updates to our translations as needed.

3.2 Addressability and Citeability

Perhaps the most critical feature of RISM Online is one will go unnoticed by most users: URLs. The structure and design of the URLs in RISM Online provide mechanisms for long-term sustainability, share-ability, and cite-ability for RISM data. RISM Online

[2]Country codes are from the International Vehicle Registration Code, https://en.wikipedia.org/wiki/International_vehicle_registration_code
[3]https://rism.online
[4]https://crowdin.com

URLs are permanent, user-friendly, compatible with most search engine crawlers, and can be maintained into the future even through changes in the software that delivers the data. RISM Online URLs are designed to function as both a locator through which you can retrieve the record—and identifiers—a string that serves as a globally unique identifier for a given thing, such as a source, a person, or an institution. The former means that, for any given address, a user can put this in their web browser and retrieve a record; the latter allows a RISM record to function within large datasets by unambiguously identifying a source, person, institution, incipit, or any other piece of data published by RISM.

While URLs in web applications are not conceptually difficult to get right, they are also, typically, the first thing to go wrong in many web-based tools. Many of the design features of RISM Online URLs are taken directly from the seminal "Cool URLs don't change" document by Tim Berners-Lee[2], but despite the general acceptance of the principles in this document as good practice, this has not previously been considered a priority when publishing the RISM data.

A core component of humanities research, or indeed any research, is the need to cite evidence of a claim. For digital resources this requirement translates to a need for long-term stability of the reference—the URL—but this often runs headlong into the easy malleability of the software that powers web resources and the convenience of the URL as a mechanism for enacting system-specific functions. There is an incongruity between how software developers use the URL to make their systems function, even following widely acceptable practices, and the need for a stable base for citation. This incongruity has an impact on the long-term stability of any digital resource. It is a very common phenomenon, for example, that new services launch with the name of the software that powers the website, or contain components that advertise some mechanism of how the software functions—a "leaky abstraction" that serves to tie the URL to the specific software delivering it[15]. When, inevitably, the underlying software application is changed, or there is a re-organization of the website, this necessitates a re-ordering of the URLs, usually breaking all the links to the resource in the process, and rendering the links in citations inoperable.

To address this in RISM Online, we designed the URLs and their functions around the core operations of our organization, around the data model we have established, and around the unified functions of both identification and retrieval, and not around the software that delivers it. The independence between the software that powers RISM Online, and the data that we publish, creates the necessary preconditions to change the underlying software without necessitating a change in the addresses of the URLs and, by extension, maintain the stability and cite-ability of the URLs into the future.

3.2.1 Types of URLs. There are two main types or URLs we publish in RISM Online. The first is the record URL, which follows the general pattern of `https://rism.online/<record-type>/<record-id>/`. From this, we have URLs such as https://rism.online/sources/605001341 or https://rism.online/people/371987.

The record type component reflects a component of the RISM data model—sources, people, institutions, and so on—while the record ID is a number assigned by Muscat at the point of record creation[5]. Combined, these components provide a unique and unambiguous identifier for the record. As a result, the construction and maintenance of the URLs presented by RISM Online are integrated into the core operations of the RISM organization, and this integration has the effect of guaranteeing stability for the foreseeable future.

The second type are search URLs. These are endpoints that accept query parameters for the purpose of searching and filtering a set of results. The main search endpoint is https://rism.online/search, but search URLs are also available on authority records, to search the sources related to that person; for example, https://rism.online/people/371987/sources is a search URL that accepts query parameters against a list of sources related to Anna Carolina Philippina Bach. While these URLs may not function as identifiers, *per se*, they are also designed to be stable and citeable, containing the necessary information to save and re-visit a search later, share it with a colleague, or cite it in a publication.

3.2.2 Content Negotiation. Early in this paper we stated that the goal for RISM Online was to present the RISM collections for both human and machine consumption. This goal requires that we make the data available through both a user interface and as machine-parseable representations.

For RISM Online, we publish only a single URL for both the human-readable record and the data, with a mechanism to vary the response from a single URL based on a supplied HTTP header. This process is known as "Content Negotiation"[6]. For humans, accessing RISM Online with a web browser, this header is supplied automatically and will return the HTML response as expected. To retrieve the data version of the same page, a script or other machine-based access method can use the `Accept` HTTP header to "negotiate" a different response type with the same URL. Providing an HTTP request header of `Accept: application/ld+json` for any URL on RISM Online will deliver a machine-readable response formatted as JSON-LD. Indeed, the human-readable view of a record in RISM Online uses this API, and the content negotiation technique, to fetch the data when a page is loaded in a web browser.

3.3 JSON-LD

JavaScript Object Notation for Linked Data (JSON-LD)[16] is a machine-readable data format that serves a dual purpose as both an easy-to-process client-server delivery format, and a mechanism for translating JSON to Linked Data. The Linked Data extensions refer to the addition of context rules that can be used to transform "ordinary" JSON into Linked Data expressions. For JSON-only tools, the Linked Data mechanisms are simply ignored. Any data not covered by the context rules is, likewise, ignored by the Linked Data tools. RISM Online delivers JSON-LD for all data responses[6].

JSON-LD provides a number of practical benefits for RISM Online. The fact that it is "just JSON" to a web browser means that the data is widely understandable by a large number of tools, including the

[5] RISM has assigned multiple types of source identifiers over the years. The identifiers assigned in the printed catalogues are no longer assigned, but may still be present in some legacy data. Identifiers for records created for the CD-ROM editions and later, however, were preserved when the data was migrated to Muscat.

[6] At the time of writing the context rules for the search data responses are experimental and have not been fully developed. As such, they are subject to change and should not be relied upon as a completely stable representation of the RISM data.

```
"label": {
  "none": [
    "Bach, Anna Carolina (1747–1804)"
  ]
}
```

Figure 1: A JSON fragment before expansion

```
"http://www.w3.org/2000/01/rdf-schema\#label":
[{
  "@value": "Bach, Anna Carolina (1747–1804)",
  "@language": "none"
}]
```

Figure 2: The expanded form of JSON-LD

JavaScript engines built into every web browser. The additional context document means that the same data can function within a Linked Data environment, enabling other methods of search and analysis on the data. Language Maps provides an interoperable method of distributing internationalized data, as both JSON and Linked Data. These will be discussed further in this section.

3.3.1 Contexts and Transformations. A JSON-LD context defines the additional statements needed to transform a simple JSON key-value statement. These statements describe the vocabularies and data structures that can be expected at a given key, and how their semantic interpretations can be expanded into URIs suitable for use in subjects, predicates, and objects.

RISM Online defines several contexts for the different types of authority records, each with the rules on how to interpret a JSON-LD response. For example, a JSON-LD response may contain the statement shown in Figure 1. Through the application of the JSON-LD context rule, this can be expanded to the form shown in Figure 2. Note the expansion of the "label" key to a fully-qualified predicate using the RDF Schema vocabulary.

This mechanism may be used in combination with the Content Negotiation method described earlier. An `Accept` header value of `application/n-triples` will return an n-triple-formatted response. This is provided through the application of a JSON-LD transformation tool on the server-side to transform the "canonical" JSON-LD representation into the n-triple-formatted response. Likewise, an `Accept` header value of `text/turtle` will return a Turtle-formatted response of the same.

3.3.2 Language Maps and Language Negotiation. Internationalization of the RISM Online API is made possible using JSON-LD Language Maps. These are data structures that provide parallel labels for data, in different languages, in a way that a client can automatically choose the label to match their desired language. RISM Online and Muscat support seven different languages: English, French, German, Italian, Polish, Portugese, and Spanish[7].

[7]An eighth language, Catalan, is available in Muscat but is not yet supported in RISM Online.

```
"createdLabel": {
  "de": ["Erstellungsdatum"],
  "en": ["Created on"],
  "es": ["Fecha de creación"],
  "fr": ["Date de création"],
  "it": ["Data di creazione"],
  "pl": ["Data utworzenia"],
  "pt": ["Criado em"]
}
```

Figure 3: An example JSON-LD Language Map

An example Language Map is given in Figure 3 showing a single field that is used as a label for the date of creation of a record.

A Language Map appears, to ordinary JSON parsers, as a simple map of parallel labels. To JSON-LD-aware parsers, however, the labels can be used in the construction of the RDF triple statements, and are expanded to provide language annotations to the RDF statements. For labels that should not be translated (for example, titles or personal names) the special value of `@none` is used as a language code. This indicates that different language values are not present, and the client should always render this value for all languages. This can be seen in the JSON-LD expansion example in Figure 2.

The translated labels are very useful for human users, but for data purposes it is often the case that only a single language, or a subset of languages, is required. For this, a custom HTTP header is available to provide language negotiation, `X-API-Accept-Language`, similar to the process for content negotiation described earlier. This header takes a comma-separated list of language codes, and will vary the content of the language maps returned by the API. Clients are expected to choose the label matching their desired language to display to their users.

3.4 National Collections

In many cases, RISM serves as the only centralized bibliography of a country's musical sources. While undoubtedly the inherent value of RISM is in the aggregate of all international contributions, it is also valuable for our contributors to identify, and indeed isolate, the collections of any given nation—particularly their own.

Previous attempts at providing this involved a dedicated Muscat instance to deliver the collections of a given country. This was provided for the United Kingdom, the Vatican ("Sistina"), Ireland, and Switzerland. Creating new Muscat installations for each of these was time consuming to set up and maintain, so using this as the basis for providing dedicated country-limited searches for all countries was not feasible. Instead, this functionality was built into RISM Online. For every country that has at least one holding institution the site provides a way to dynamically limit all searches to sources held by that country[8].

[8]Once activated, the dedicated search interface for the people authorities is removed, since a person is not limited to a specific country, and the RISM data was not sufficiently expressive to support these restrictions. For example, the record for Ludwig van Beethoven has relationships to Bonn and Wien, but the data to tie these cities to modern countries, and thus to national collections, is not captured by RISM.

Restricting RISM Online to a National Collection can also be done by URL parameters, so a national working group can link directly to country-limited search interface from their own websites. It also works with the Search API, so it is possible to retrieve the JSON-LD response and present a custom search interface. To use this, the URL parameter `nc` is supplied with the country code from the RISM sigla. So `nc=D` would restrict the site to German sources and institutions, while `nc=CDN` would do the same for Canadian ones.

3.5 Incipits

Attaching musical incipits to cataloguing records has long been a part of RISM cataloguing, enabled by the development of the Plaine and Easie Code (PAEC) for encoding the incipits using simple alphanumeric characters available on any typewriter. Although PAEC was ostensibly "music notation," in the past it took a specialist and a bit of imagination to make sense of these textual encodings as music notation, since visualizing them as recognizable music notation was a relatively arduous process. Today, however, the Muscat tool integrates the Verovio notation renderer[9], and cataloguers can see their incipits rendered as music notation as they type. Verovio also features an extensive set of validation rules ensuring the incipits follow the rules of the PAEC scheme.

The primary purpose of capturing incipits is to help describe musical content, at the notation level, where other pieces of information, such as title, composer, key, or tempo, are not sufficient to fully describe the work[10]. Incipits are used to disambiguate and to collate; to tell two musical works apart despite having the same title, or to find the same piece of music despite having different titles. They can also be used as the basis for notation search engine, allowing users to query for musical sources or works based on a melodic fragment.

3.5.1 Incipit Search. There are a number of notable search engines built using the RISM incipit data. The RISM Themefinder website[11] [11, 14] was one of the first and allows users to search an older copy of the RISM incipits from source collections in the United States. The RISM Search by Utrecht University[12] [19] uses an advanced analysis of the RISM incipits, using several custom C++ components embedded in a Solr plugin. The RISM Incipit Search[13] [13] by Jacek Salomon uses Locality Sensitive Hashing (LSH) approach with a custom set of ASP tools for indexing and a MySQL database. The RISM OPAC[14] has had music incipit search since 2011. The current version uses *n*-gram indexes of intervals or exact pitch and octave (without considering enharmonic equivalents) to match a user's query[1].

These projects provided important prior art for designing our own incipit search system. However, many of these prior solutions depend on bespoke software components that are not supported, or on systems that were not part of our existing systems architecture. We identified seven criteria to guide the development of our incipit search:

- It needed to work with our existing metadata search setup in Solr[15] and not depend on complex, custom, or unsupported tools so that we could deliver a stable service for our users;
- It needed to provide very fast indexing speeds, in combination with the metadata indexing, so that we could meet our goal of providing daily updates from Muscat;
- It needed to provide fast results from queries to meet our user experience goal of providing a responsive service;
- It needed to support result highlighting so that users could see immediately why a result was returned, and what the search system thought that it matched;
- It needed to be "encoding independent"—that is, two incipits of the same piece, encoded with a different sequence of characters, would still match;
- It needed to provide options on how the queries would be evaluated; either pitch invariant (the query would match the target documents regardless of the transposition), pitch dependent (the exact pitches would need to be used to qualify as a match), or contour (the query would match the incipits as long as the direction of the intervals—up, down, or same—matched);
- It needed to work with PAEC as a query language, so that music notation queries could be expressed at varying levels of complexity.

The starting point for our incipit search was to replicate the work done by Salamon for his incipit search engine. With his help we started with reproducing his solution in MySQL, for which he supplied a database export so that we could observe the software's operation. We then re-implemented this version using the built-in LSH functionality in Solr. While these results were promising, the results could sometimes produce hashes that matched computationally, but not musically. Instead, a simpler method was developed that provided better results.

Our incipit indexing uses feature extraction built into Verovio. This loads the PAEC of the incipit and produces several representations of the incipit: arrays of Chromatic and Diatonic intervals and pitches, and the gross and fine contour of the incipit. Verovio assigns a unique ID to every note in the PAEC input. This unique ID is randomly generated, but the seed for the random generator is the CRC hash of the PAEC input. Thus, two PAEC encodings that are the same will produce the exact same random IDs for each note, but two that are different, even by a single character, will generate completely different IDs. (This will be an important feature later in the search process.) For pairwise features, such as interval or contour, a two-dimensional array of these IDs is produced, identifying the pairs of notes. The features from Verovio are stored in Solr as multivalued lists, and the two-dimensional ID arrays are stored as JSON strings. The original PAEC of the incipit is also stored in the

[9] https://verovio.org

[10] "The thematic index derives its superiority over non-thematic lists because it can not only arrange a body of music in a systematic order, but it provides, at the same time, positive identification in a minimum of space and symbols. It does so by the use of the "incipit," or musical citation of the opening notes. For most music, an incipit of no more than a dozen pitches is required. When rhythmic values accompany the pitches, the incipit's "uniqueness quotient" is astonishingly high.

A musical work, printed or manuscript, may be identified by composer, title, opus number, key, instrumentation, movement headings, first line of text, date, publisher, dedicatee, plate number, etc. No one of these, indeed no combination of these, can provide as certain an identification as an incipit"[4], p. 408

[11] http://rism.themefinder.org

[12] https://www.projects.science.uu.nl/monochord/risma2/query/db

[13] http://musicalsources.org

[14] https://opac.rism.info

[15] https://solr.apache.org

Solr document. Experiments with storing the SVG directly in Solr resulted in an unreasonably large search index size.

Each set of features stored in Solr as whitespace-delimited strings, and each string is analyzed by the built-in Shingle filter. A shingle is like an *n*-gram, but it operates on the level of words (i.e., tokens) instead of individual characters. This distinction is important, since intervals and pitches may contain more than one character that is integral to the full token, e.g., `-1` or `bB`. The shingle filter produces token shingles of length three, capturing the sequence as a set of overlapping 3-length sub-sequences.

When a user searches for an incipit, their search query is interpreted as PAEC, allowing them to express both simple and complex notation queries. This query is run through the same Verovio feature extractor, producing a set of features that can be matched against the documents in Solr. The extracted features are analyzed as shingles by the Solr query analyzer. The search is then performed on the shingle representations for a selected query type (intervals, exact pitches, contour), where Solr attempts to match the shingle sequences with the best overlap, scoring the matches accordingly. The Boolean Similarity Factory in Solr provided the best results for identifying matches. This similarity measure gives a score based on a simple count of token matches between the query and the document.

Once the search has narrowed down the number of documents, they are ranked so that they can be sorted with the most "relevant" results first. In this case, relevancy is measured as the greatest boolean similarly overlap between the query and a target document. To rank the results we compute the Jaccard similarity[5, 9] of each incipit, and order them from most similar to least similar:

$$\frac{DS}{((I+Q)-DS))}$$

Where DS is the document score calculated by the Solr query, I is the length of the document incipit, and Q is the length of the query incipit. The results are then sorted by this score. Since we use the boolean similarity measure, the document score is the length of the intersection, as measured by Solr, while the incipit and query lengths are the union.

At this point in the process we have a ranked list of incipits that match a given query, so we know that there must be some sequence of tokens within this list of results that overlap with the query. We then need to provide the user some visual indication of where, in any given incipit, the notes in their query matched the target.

We use the Longest Contiguous Subsequence (LCS), as implemented in the Python `difflib` module[16][12], between the stored incipit features and the features extracted from the query. This provides us with start and ending index positions of the overlapping sequences of tokens. The original PAEC for this document is loaded into Verovio to render as SVG. Since we use the CRC of the input as the seed for the random ID generator, the IDs in the newly loaded file, and the list of IDs stored in Solr, are the same. The start and end index of the list of IDs representing the LCS are then extracted by their index and used as CSS style selectors injected into the SVG to highlight each matching pair of notes by their ID. This is presented to the user as rendered music notation where they can clearly see

[16]https://docs.python.org/3/library/difflib.html

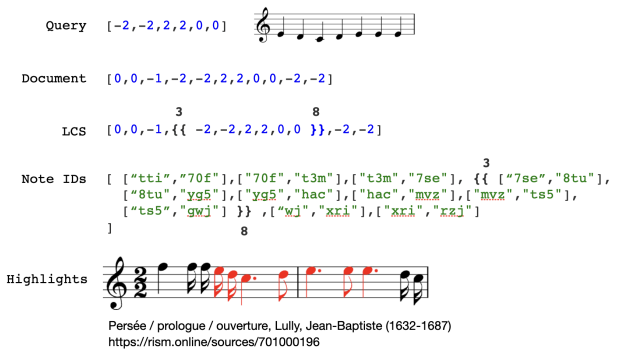


Figure 4: Highlighting the Longest Contiguous Subsequence

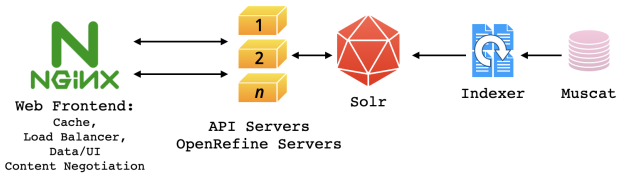


Figure 5: RISM Online Service Architecture

their query sequence highlighted in the list of matches produced by Solr. This process is illustrated in Figure 4.

3.6 Service Delivery

The underlying technical architecture of RISM Online is a set of micro-services, spread over several virtual machines. An overview of the RISM Online architecture is shown in Figure 5.

Most of the individual components of the RISM Online service delivery setup are available under an Open Source license. A list of the software repositories for the tools mentioned in this section is given in Appendix A.

3.6.1 Indexing and Search. At the center of RISM Online is the Solr search engine, which serves as both a search engine and a document store. An indexing tool processes the data from the Muscat database, extracting data from MARC21 and sends it to Solr for indexing. No other database needed for the site to operate. The indexer runs once a day and currently takes approximately 30 minutes to complete a full re-index on our production server (8 virtual CPUs, 15 GB RAM). The indexer is implemented in Python, and uses parallel processing to distribute the indexing tasks over all available CPUs on the machine. Almost 6 million Solr documents are created in the indexing process, representing all source descriptions, authority records, holdings, incipits, and others.

3.6.2 Data Delivery. The data for RISM Online is delivered through the API Servers. The primary role of this component is to interact with the Solr search engine, and transform data from the internal Solr document structure to the public JSON-LD representation. The API Servers are designed to be run in parallel behind a load balancer, allowing for horizontal scaling of the service as needed.

We use NginX[17] for our front-end web server. NginX plays several roles in the service delivery of RISM Online. It acts as a load-balancer, distributing traffic between two instances of the API Servers. It is also a front-end cache, intercepting requests for data and delivering a cached version, if available. Finally, it handles content negotiation, directing requests for the HTML or Data representations to the appropriate handler.

3.6.3 User Interfaces. The User Interface component is a single-page JavaScript application that handles all user-facing elements and translates the JSON-LD data from the API to human-readable web pages. It is developed using the Elm language[18], which is compiled to an optimized JavaScript bundle.

In addition to the main RISM Online interface application, we have developed two complementary user-facing applications. The first is "Copperplate", a music notation viewer application that can accept a URL to an encoded notation file and render it to the browser using Verovio. Copperplate is used in RISM Online to render linked music notation files in the records, allowing users to see the notation in their browser instead of downloading the file and opening it in another music notation viewer.

The second standalone application is "Apero", an API data viewer. Apero serves as a bridge between the user interface and the data APIs. It presents a user interface for adjusting the HTTP request parameters, allowing users to see the results of content negotiation in the browser and view the data of their choices.

3.6.4 OpenRefine Reconciliation Server. An OpenRefine Reconciliation API service[19] is available for the RISM data. This was initially built to help perform DIAMM integration, but is now available as a public service for anyone to use[20]. As a public service the OpenRefine infrastructure can help us integrate additional databases into our own system, or can help others integrate RISM Online source and authority descriptions into their projects.

4 DIGITAL IMAGE ARCHIVE OF MEDIEVAL MUSIC

The Digital Image Archive of Medieval Music (DIAMM)[21] is the preeminent scholarly source for online descriptions of Medieval and Renaissance polyphonic music manuscripts. DIAMM contains most of the RISM Series B/IV descriptions. The DIAMM project independently transcribed most of the source descriptions with permission and hosts the most complete online version of Series B/IV descriptions, as well as descriptions of newly found sources and inventories created since the B/IV catalogue was printed.

For people looking for Medieval and Renaissance manuscript descriptions, the lack of online availability in RISM Online was a constant source of confusion, particularly for those who are used to consulting the printed catalogues. To help our users find these sources, we have partnered with the DIAMM project to make their source descriptions, as well as person and institution authorities, available through RISM Online. The goal with this effort is to make the source descriptions find-able on a RISM platform. With this integration, over 3,700 source, 2,500 people, and 700 institution records from the DIAMM project are now made available for finding in RISM Online.

One of our biggest concerns in this integration was that we did not simply want to harvest the DIAMM records into RISM Online, and in the process create duplicate records. This is a common method of providing "federated" search, but can also lead to confusion as users must find more than one result to get the full picture. Instead, we developed two paths for integration between the two sets of records. The first is between records that exist in both RISM and DIAMM. For these records, the RISM record is preferred, but a link is added to the record for the corresponding entry in DIAMM. Aligning these records is done by storing the RISM Online identifier in the DIAMM record, a process that was done using OpenRefine[22] and the RISM Reconciliation service described previously. The second path is for records that are only in DIAMM. In this case the record description is harvested from DIAMM and integrated into the Solr search engine as independent results. These results are shown with the RISM records for searches with a DIAMM logo identifying as such. While we display a preview of the record in the search results, the "Full Record" link points users back to the DIAMM website, where they have access to more detail about the record, including source inventories, images or links to external resources.

The DIAMM integration has filled a significant gap in the online availability of RISM descriptions for Medieval and Renaissance polyphony. In the future we will be exploring further collaborations with other projects, and examining whether similar methods of integration is suitable and desirable to our user community.

5 FUTURE WORK

RISM Online is an infrastructure project for the global RISM community, and will continue to improve. This section will describe some of the new features that are currently being worked on and will make their way to the public interface.

5.1 Linked Data

RISM Online has been built with Linked Data at the top of the design priorities. The focus on URL and URI design, as noted earlier in this paper, was fundamental to establishing it as a resource within the Linked Data environment, and a necessary step to providing this as a service in the future.

We are seeing a growing need to expand beyond simply making RISM data available. Our users are asking for more tools to do tasks related to data *analysis*—asking questions of the data that go beyond simple results retrieval. One component of this analysis is making the RISM data available in a graph representation, to better search and analyse the complex relationships between records.

Many of the queries our users struggle with asking are those that are not suitable for simple search interfaces, and are instead those that require graph-based queries, traversing multiple record types and statements to fully answer their questions. Questions such as "What person is the most popular dedicatee in all the RISM sources?" or "Visualize the use of major and minor keys over time"

[17] https://nginx.org
[18] https://elm-lang.org
[19] https://openrefine.org/docs/technical-reference/reconciliation-api
[20] https://rism.online/docs/reconciliation/introduction/
[21] https://www.diamm.ac.uk
[22] https://openrefine.org

are surprisingly difficult to answer in the RISM Online UI, but can be answered by structuring the queries in a Linked Data context.

While the JSON-LD expansion into RDF has worked as an extended proof-of-concept, it needs further enhancement to more completely describe the RISM data as Linked Data, and make it suitable for such analysis tasks. When this is suitably developed, we will be making it available through a SPARQL query interface, and our context documents will provide users with a standardized and tested means of transforming RISM data into Linked Data.

5.2 New Developments for RISM

The ability to control and experiment with our own data means we are better able to adapt the RISM project itself to meet demands from our users. We are currently working on building a Musical Works authority, and will make this data available in RISM Online through a dedicated search interface.

RISM Online is also playing a central role in our own internal efforts to validate and correct legacy data. By building better search and analysis tools for our users, we are also ourselves gaining new insights into the contents of the RISM data and, in particular, where this data can be improved and made more consistent and expressive.

6 CONCLUSION

Over the past 70 years, RISM has created a music source catalogue that is a core part of music source scholarship. As a "side-effect," the project has also created a data resource unlike any other in musicology. By making the complete dataset accessible through dedicated APIs and stable URIs, RISM Online transforms the RISM dataset into a central point for new digital musicology endeavours.

Our hope is that RISM Online drives new methods for musical data searching and analysis. The incipit search system detailed in this paper is the largest and fastest of its kind, updated daily and built using well-supported commodity tools. Extensive use of JSON-LD opens the door to expanded access and analysis capabilities through Linked Data tools, and our hope is that it will continue to be a valuable tool for our global community of users.

ACKNOWLEDGMENTS

Thank you to Alastair Porter for providing the Jaccard index as a Solr ranking method and the use of the Boolean Similarity method.

REFERENCES

[1] Sanu Pulimootil Achankunju. 2018. Music Search Engine from Noisy OMR Data. In *1st International Workshop on Reading Music Systems*, Jorge Calvo-Zaragoza, Jan Hajič jr., and Alexander Pacha (Eds.). Paris, France, 7–8.

[2] Tim Berners-Lee. 1998. *Cool URIs don't change.* W3C. https://www.w3.org/Provider/Style/URI.html

[3] Barry S. Brook. 1965. The Simplified "Plaine and Easie Code System" for Notating Music : A Proposal for International Adoption. *Fontes Artis Musicae* 12, 2/3 (1965), 156–160. http://www.jstor.org/stable/23504707

[4] Barry S. Brook. 1973. A Tale of Thematic Catalogues. *Notes* 29, 3 (1973), 407–415. https://doi.org/10.2307/896416

[5] Tim Crawford, David Lewis, and Alastair Porter. 2023. Exploring early vocal music and its lute arrangements: Using F-TEMPO as a musicological tool. In *Proceedings of the 10th International Conference on Digital Libraries for Musicology*. Association for Computing Machinery, Milan, Italy, 77–81.

[6] M. Nottingham Fielding, Roy and J. Reschke. 2022. *HTTP Semantics.* Internet Engineering Task Force (IETF). https://datatracker.ietf.org/doc/html/rfc9110

[7] Andrew Hankinson and Jennifer Ward. 2023. The Past, Present, and Future of the Plaine and Easie Code. In *The Joint MEI and TEI Conference*. MEI/TEI, Paderborn, Germany.

[8] Jennifer Hughes. 2008. *Analytical Cataloging: Past, Present, and Future Implementations.* http://jhughes.wikidot.com/analytical-cataloging

[9] Paul Jaccard. 1901. Étude comparative de la distribution florale dans une portion des Alpes et du Jura. *Bulletin de la Société Vaudoise des Sciences Naturelles* 37, 142 (1901), 547.

[10] Laurent Pugin. 2020. *History of the Muscat Project.* RISM Digital Center. http://muscat-project.org/history.html

[11] Laurent Pugin and Rodolfo Zitellini. 2013. Instrument distribution and music notation search for enhancing bibliographic music score retrieval. In *Proceedings of the 13th ACM/IEEE-CS Joint Conference on Digital Libraries* (Indianapolis, Indiana) *(JCDL '13)*. Association for Computing Machinery, New York, NY, USA, 195–198. https://doi.org/10.1145/2467696.2467728

[12] John W Ratcliff and David Metzener. 1988. Pattern matching: The gestalt approach. *Dr. Dobb's Journal* 141 (1988), 355–358.

[13] Jacek Salamon. 2018. *Searching Music Incipits in Metric Space with Locality-Sensitive Hashing.* https://www.codeproject.com/Articles/1268315/Searching-music-incipits-in-metric-space-with-loca

[14] Craig Sapp, Yi-Wen Liu, and Eleanor Selfridge-Field. 2004. Search Effectiveness Measures for Symbolic Music Queries in Very Large Databases. In *Proceedings of the 5th International Conference on Music Information Retrieval*. ISMIR, Barcelona, Spain.

[15] Joel Spolsky. 2002. *The Law of Leaky Abstractions.* https://www.joelonsoftware.com/2002/11/11/the-law-of-leaky-abstractions/

[16] Manu Sporny, Dave Longley, Gregg Kellogg, Markus Lanthaler, Pierre-Antoine Champin, and Niklas Lindström. 2020. *JSON-LD 1.1: A JSON-based Serialization for Linked Data.* World Wide Web Consortium (W3C). https://www.w3.org/TR/json-ld/

[17] Roy Tennant. 2010. *MARC Must Die.* Library Journal. https://www.libraryjournal.com/story/marc-must-die

[18] RISM Zentraldirektion. 1997. RISM In Internet. *Info RISM* 8 (1997), 8–11. https://rism.info/resources-old-website/site-content/Info-RISM/InfoRISM_08.pdf

[19] Rodolfo Zitellini, Geert-Jan Giezeman, Frans Wiering, and Laurent Pugin. 2018. Incipit Melodic Similarity Matching in Muscat. In *Proceedings of the 19th International Conference on Music Information Retrieval*. ISMIR, Paris, France.

A OPEN SOURCE REPOSITORIES

This is a list of the Open Source projects developed to support RISM Online. (The original internal name for RISM Online was "MuscatPlus", which is why many of the projects still use this name.)

Data Server The Python server that runs the data API. https://github.com/rism-digital/muscatplus_server

Indexer Converts Muscat (and other data) to the RISM Online Solr index https://github.com/rism-digital/muscatplus_indexer

OpenRefine Reconciliation Server https://github.com/rism-digital/muscatplus_reconciliation/

User Interface The browser-based client application https://github.com/rism-digital/muscatplus_ui

Solr configuration Our Solr configuration https://github.com/rism-digital/muscatplus_solr

Copperplate Notation Viewer A web-based notation viewer built around Verovio https://github.com/rism-digital/copperplate

Apero API Viewer A simple browser-based API viewer https://github.com/rism-digital/apero

small-asc Solr Client An asynchronous Solr client https://github.com/rism-digital/small-asc

Ypres A fork of the Serpy serializer library, modified to support asynchronous Python. https://github.com/rism-digital/ypres

EDTF A fork of the Extended Date-Time Format Library with performance enhancements. https://github.com/rism-digital/python-edtf

Verovio The feature extractor functionality was developed to support the incipit search. https://github.com/rism-digital/verovio